More on KL Divergence

$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$

- The KL divergence measures the expected number of extra bits required to code samples from *p*(*x*) (“true” distribution) when using a code based on *q*(*x*), which represents a theory, model, description, or approximation of *p*(*x*)
- The KL divergence is not a distance measure, not a metric: asymmetric, not satisfy triangular inequality ($D_{KL}(P||Q)$ does not equal $D_{KL}(Q||P)$)
- In applications, *P* typically represents the "true" distribution of data, observations, or a precisely calculated theoretical distribution, while *Q* typically represents a theory, model, description, or approximation of *P*.
- The Kullback–Leibler divergence from *Q* to *P*, denoted $D_{KL}(P||Q)$, is a measure of the information gained when one revises one's beliefs from the prior probability distribution *Q* to the posterior probability distribution *P*. In other words, it is the amount of information lost when *Q* is used to approximate *P*.
- The KL divergence is sometimes also called the information gain achieved if *P* is used instead of *Q*. It is also called the relative entropy of *P* with respect to *Q*.

Subtlety at Computing the KL Divergence

- *Base on the formula,* $D_{KL}(P,Q) \geq 0$ and $D_{KL}(P \,||\, Q) = 0$ if and only if $P = Q$
- How about when p = 0 or q = 0?

$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$

  - $lim_{p \to 0}\, p \log p = 0$
  - when p != 0 but $q = 0$, $D_{KL}(p \,||\, q)$ is defined as ∞, i.e., if one event e is possible (*i.e.*, $p(e) > 0$), and the other predicts it is absolutely impossible (*i.e.*, $q(e) = 0$), then the two distributions are absolutely different
- However, in practice, P and Q are derived from frequency distributions, not counting the possibility of unseen events. Thus *smoothing* is needed
- Example: P : (a : 3/5, b : 1/5, c : 1/5). Q : (a : 5/9, b : 3/9, d : 1/9)
  - need to introduce a small constant ϵ, e.g., $\epsilon = 10^{-3}$
  - The sample set observed in P, $SP = \{a, b, c\}$, $SQ = \{a, b, d\}$, $SU = \{a, b, c, d\}$
  - Smoothing, add missing symbols to each distribution, with probability ϵ
  - P' : (a : $3/5 - \epsilon/3$, b : $1/5 - \epsilon/3$, c : $1/5 - \epsilon/3$, d : ϵ)
  - Q' : (a : $5/9 - \epsilon/3$, b : $3/9 - \epsilon/3$, c : ϵ, d : $1/9 - \epsilon/3$)
  - $D_{KL}(P' \,||\, Q')$ can then be computed easily